

Constructed local graph

$$\mathcal{G}_a = (\mathcal{V}_a, \mathcal{E}_a)$$

node and edge features
fixed candidate relations
which relations are active
which nodes are visible
which nodes are controlled

Shared graph encoder

Node projection

Linear $d_v \rightarrow d_h$
ReLU, then LayerNorm

 $\mathbf{H}^{(0)}$

Active-edge filtering

keep only the candidate relations
that are electrically active
at this step

$K \times$ **message-passing layers**

configurable graph encoder

ReLU, then LayerNorm

$$\mathbf{H}^{(0)} \rightarrow \mathbf{H}^{(K)}$$

 $\mathbf{H}^{(K)}$

Readout

average $\mathbf{h}^{(K)}$ over the
visible nodes $\Rightarrow \bar{\mathbf{h}}_a$

controlled_readout: controlled nodes only

 $\bar{\mathbf{h}}_a$

Graph projection

Linear $d_h \rightarrow d_z$
ReLU $\Rightarrow \mathbf{z}_a$

 $\mathbf{z}_a \in \mathbb{R}^{d_z}$

Agent-specific action head

Actor MLP

configurable hidden layers + ReLU
Linear $d_\pi \rightarrow |\mathcal{A}_a|$

Action logits

$$\ell_a \in \mathbb{R}^{|\mathcal{A}_a|}$$